CL30

Composite Lifting-Surface Structural Sizing & Optimization

Materials Database Report

Constituent plies · Laminate stiffness · Laminate strength

22	21	10	2026-05-22
Laminates	Fiber/matrix plies	Core materials	Date

Engineering constants and strength allowables (MPa)

Material	t (mm)	E1 (MPa)	E2 (MPa)	G12 (MPa)	ν12	Xt (MPa)	Xc (MPa)	Yt (MPa)	Yc (MPa)	S (MPa)
AS4--3501-6-UD	0.125	142,000	10,300	5,700	0.270	2,280	1,440	57	228	71
Boron--Epoxy-B4--5505	0.125	204,000	18,500	5,590	0.230	1,260	2,500	61	202	67
Boron--Epoxy-B56--5505	0.125	201,000	21,700	5,400	0.170	1,380	1,600	56	125	62
CFRP-AS--3501	0.125	138,000	8,960	7,100	0.300	1,447	1,447	52	206	93
CFRP-AS4--3501-6	0.125	147,000	10,300	7,000	0.270	2,280	1,725	57	228	76
CFRP-AS4--APC2-PEEK	0.125	138,000	8,700	5,000	0.280	2,060	1,100	78	196	157
CFRP-IM6G--3501-6	0.125	169,000	9,000	6,500	0.310	2,240	1,680	46	215	73
CFRP-IM7--977-3	0.125	190,000	9,900	7,800	0.350	3,250	1,590	62	200	75
CFRP-ModI--WRD9371-PI	0.125	216,000	5,000	4,500	0.250	807	655	15	71	22
CFRP-T300--5208	0.125	181,000	10,300	7,170	0.280	1,500	1,500	40	246	68
CFabric--Epoxy-AGP370-5H--3501-6S	0.300	77,000	75,000	6,500	0.060	963	900	856	900	71
E-Glass--Epoxy	0.250	41,000	10,400	4,300	0.280	1,140	620	39	128	89
GFRP-Scotchply-1002	0.125	38,600	8,270	4,140	0.260	1,062	610	31	118	72
GrFRP-GY70--934	0.125	294,000	6,400	4,900	0.230	985	690	29	98	49
Kevlar49--Epoxy	0.125	80,000	5,500	2,200	0.340	1,400	335	30	158	49
Kevlar49--Epoxy-TH	0.125	76,000	5,500	2,300	0.340	1,400	235	12	53	34
Kevlar49-Fabric--Epoxy-K120--M10.2	0.300	29,000	29,000	18,000	0.050	369	129	369	129	113
S-Glass--Epoxy	0.250	45,000	11,000	4,500	0.290	1,725	690	49	158	70
WGlass--Epoxy-120--3501-6	0.300	27,500	26,700	5,500	0.140	435	435	386	386	55
WGlass--Epoxy-7781--5245C	0.300	29,700	29,700	5,300	0.170	367	549	367	549	97
WGlass--Epoxy-M10E--3783	0.300	24,500	23,800	4,700	0.110	433	377	386	335	84

Density and classification of foam and honeycomb core materials

Material	t (mm)	ρ (kg/m³)	Type
Al-Honeycomb-3pcf	12.7	48	Aluminium Honeycomb
Al-Honeycomb-PAMG-5052	5.0	130	Aluminium Honeycomb
Balsa-CK57	10.0	150	Balsa Wood
Divinycell-H100	5.0	100	PVC Structural Foam
Divinycell-H160	5.0	160	PVC Structural Foam
Divinycell-H250	10.0	250	PVC Structural Foam
Divinycell-H80	5.0	80	PVC Structural Foam
Foam-Honeycomb-Style-20	10.0	128	Aluminium Honeycomb
Polyurethane-FR-3708	10.0	128	Polymeric Foam
Rohacell-71IG	10.0	0	PMI Rigid Foam

In-plane engineering constants derived from CLT (A-matrix)

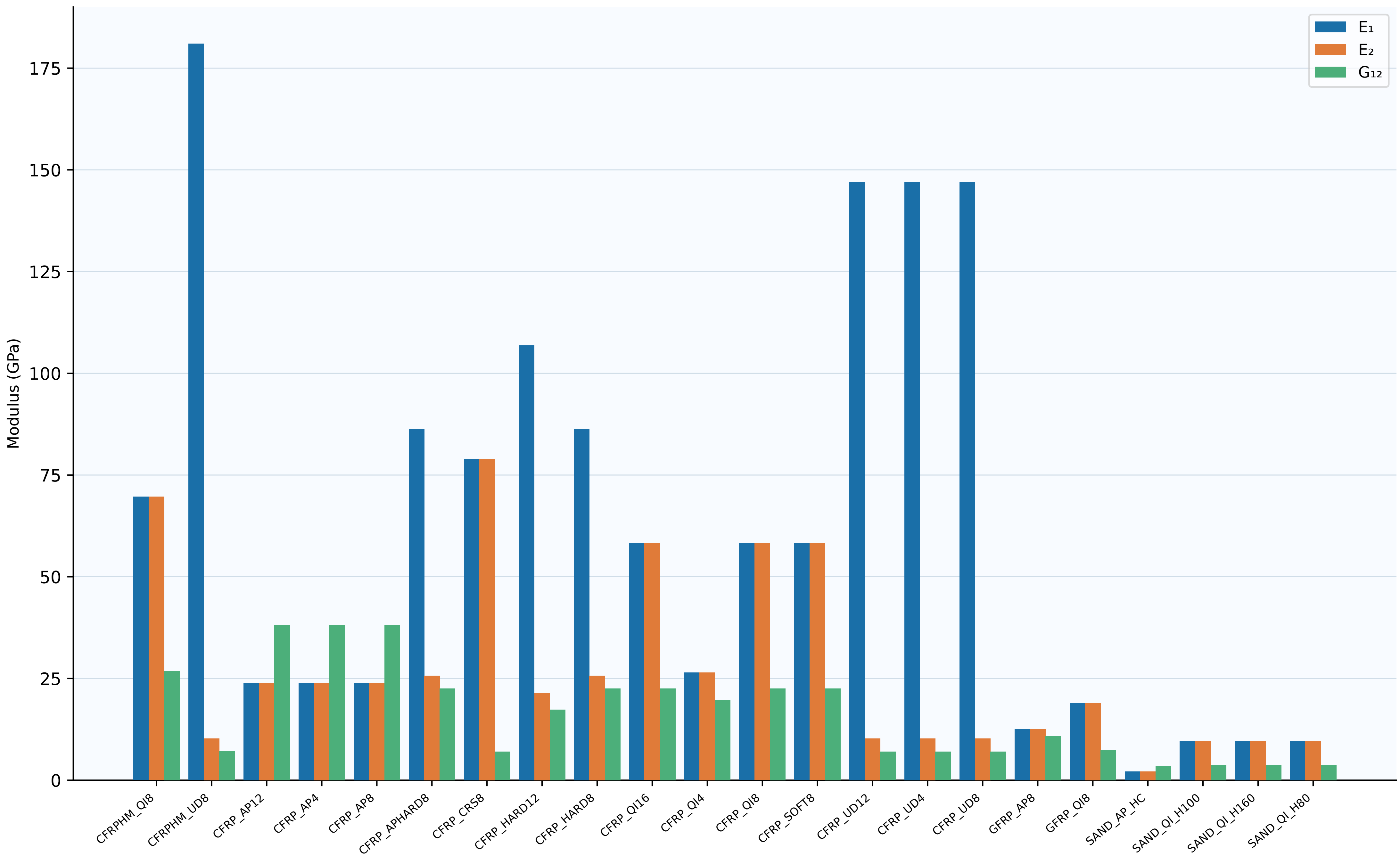
Laminate	Lay-up	n	t (mm)	ρ (kg/m³)	GSM (g/m²)	E1 (MPa)	E2 (MPa)	G12 (MPa)	ν12
CFRP_AP12	[45,-45,45,-45,45,-45]_s	12	1.50	1600.0	2400	23,910	23,910	38,129	0.7079
CFRP_AP4	[45,-45]_s	4	0.50	1600.0	800	23,910	23,910	38,129	0.7079
CFRP_AP8	[45,-45,45,-45]_s	8	1.00	1600.0	1600	23,910	23,910	38,129	0.7079
CFRP_APHARD8	[45,-45,0_2]_s	8	1.00	1600.0	1600	86,271	25,693	22,565	0.6301
CFRP_CRS8	[0,90,0,90]_s	8	1.00	1600.0	1600	78,955	78,955	7,000	0.0354
CFRP_HARD12	[0_4,45,-45]_s	12	1.50	1600.0	2400	106,894	21,361	17,376	0.5758
CFRP_HARD8	[0_2,45,-45]_s	8	1.00	1600.0	1600	86,271	25,693	22,565	0.6301
CFRP_QI16	[0,45,-45,90,0,45,-45,90]_s	16	2.00	1600.0	3200	58,180	58,180	22,565	0.2892
CFRP_QI4	[0,45,-45,90]	4	0.50	1600.0	800	26,522	26,522	19,633	0.3776
CFRP_QI8	[0,45,-45,90]_s	8	1.00	1600.0	1600	58,180	58,180	22,565	0.2892
CFRP_SOFT8	[45,-45,0,90]_s	8	1.00	1600.0	1600	58,180	58,180	22,565	0.2892
CFRP_UD12	[0_6]_s	12	1.50	1600.0	2400	147,000	10,300	7,000	0.2700
CFRP_UD4	[0_2]_s	4	0.50	1600.0	800	147,000	10,300	7,000	0.2700
CFRP_UD8	[0_4]_s	8	1.00	1600.0	1600	147,000	10,300	7,000	0.2700
CFRPHM_QI8	[0,45,-45,90]_s	8	1.00	1600.0	1600	69,676	69,676	26,880	0.2960
CFRPHM_UD8	[0_4]_s	8	1.00	1600.0	1600	181,000	10,300	7,170	0.2800
GFRP_AP8	[45,-45,45,-45]_s	8	1.00	1800.0	1800	12,556	12,556	10,799	0.5164
GFRP_QI8	[0,45,-45,90]_s	8	1.00	1800.0	1800	18,965	18,965	7,469	0.2695
SAND_AP_HC	[45,-45,0,-45,45]	5	5.50	263.6	1450	2,174	2,174	3,466	0.7079
SAND_QI_H100	[0,45,-45,90,0,90,-45,45,0]	9	6.00	350.0	2100	9,697	9,697	3,761	0.2892
SAND_QI_H160	[0,45,-45,90,0,90,-45,45,0]	9	6.00	400.0	2400	9,697	9,697	3,761	0.2892
SAND_QI_H80	[0,45,-45,90,0,90,-45,45,0]	9	6.00	333.3	2000	9,697	9,697	3,761	0.2892

CFRP (AS4/3501-6) CFRP High-Modulus (T300/5208) GFRP (Scotchply-1002) Sandwich (CFRP face-sheets)

Extensional (A) and bending (D) stiffness terms per unit width (CLT)

Laminate	A11 (N/mm)	A22 (N/mm)	A66 (N/mm)	D11 (N·mm)	D22 (N·mm)	D66 (N·mm)	Coupled?
CFRPHM_QI8	76,368	76,368	26,880	10,691	2,653	1,932	No
CFRPHM_UD8	181,811	10,346	7,170	15,151	862	598	No
CFRP_AP12	71,887	71,887	57,194	13,479	13,479	10,724	No
CFRP_AP4	23,962	23,962	19,065	499	499	397	No
CFRP_AP8	47,925	47,925	38,129	3,994	3,994	3,177	No
CFRP_APHARD8	97,840	29,139	22,565	5,034	3,602	2,853	No
CFRP_CRS8	79,054	79,054	7,000	8,735	4,441	583	No
CFRP_HARD12	171,717	34,315	26,065	40,516	3,303	2,293	No
CFRP_HARD8	97,840	29,139	22,565	11,273	1,254	908	No
CFRP_QI16	126,978	126,978	45,129	55,694	29,931	14,557	No
CFRP_QI4	31,745	31,745	11,282	783	783	113	Yes
CFRP_QI8	63,489	63,489	22,565	8,754	2,314	1,637	No
CFRP_SOFT8	63,489	63,489	22,565	4,855	3,781	2,853	No
CFRP_UD12	221,632	15,529	10,500	41,556	2,912	1,969	No
CFRP_UD4	73,877	5,176	3,500	1,539	108	73	No
CFRP_UD8	147,755	10,353	7,000	12,313	863	583	No
GFRP_AP8	17,121	17,121	10,799	1,427	1,427	900	No
GFRP_QI8	20,450	20,450	7,469	2,477	1,035	570	No
SAND_AP_HC	23,962	23,962	19,065	165,240	165,240	131,467	No
SAND_QI_H100	63,489	63,489	22,565	517,127	446,279	170,872	No
SAND_QI_H160	63,489	63,489	22,565	517,127	446,279	170,872	No
SAND_QI_H80	63,489	63,489	22,565	517,127	446,279	170,872	No

In-plane moduli E_1, E_2 and G_{12} (GPa) — all laminates



GSM (g/m²) vs laminate thickness — sized by density